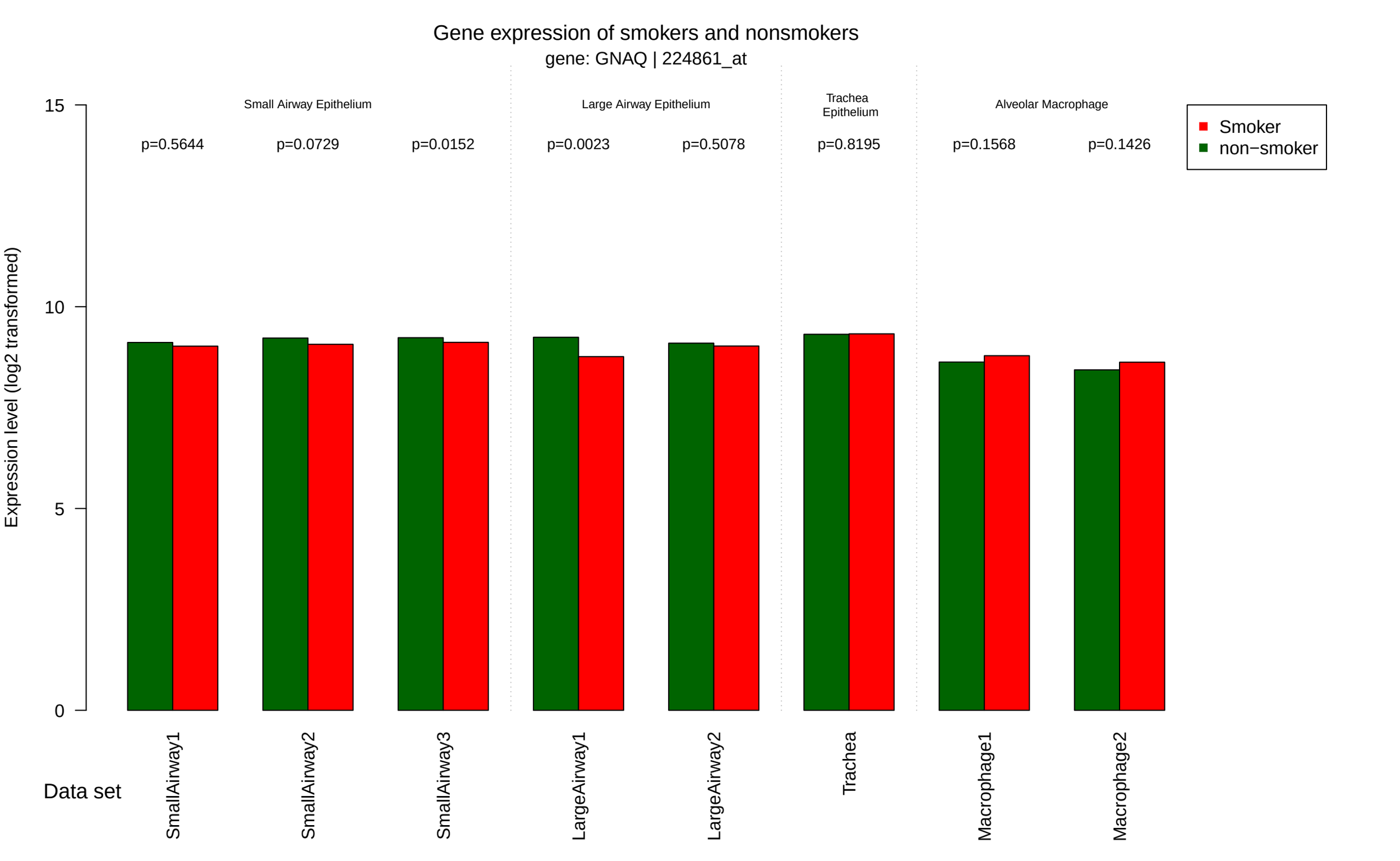
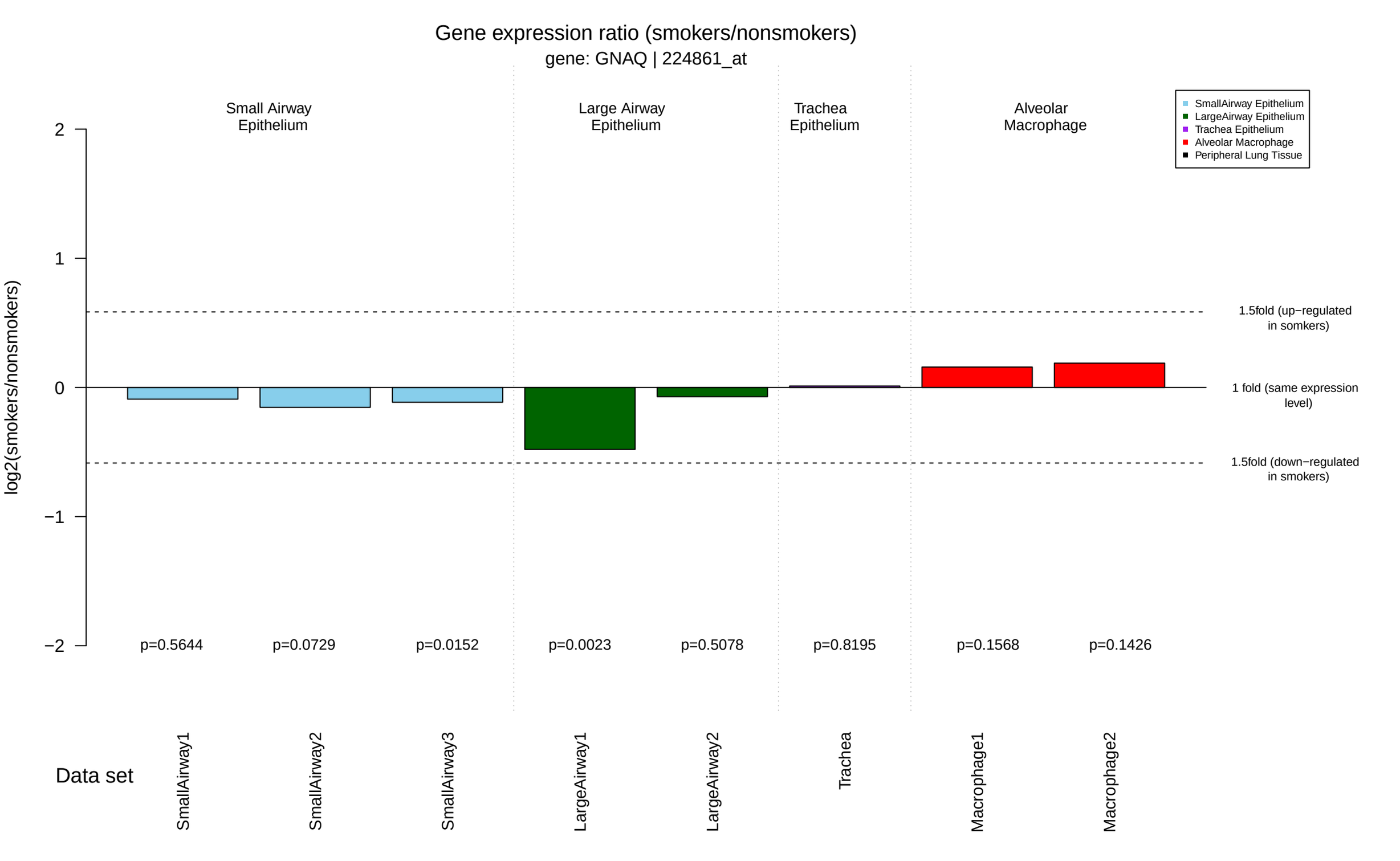
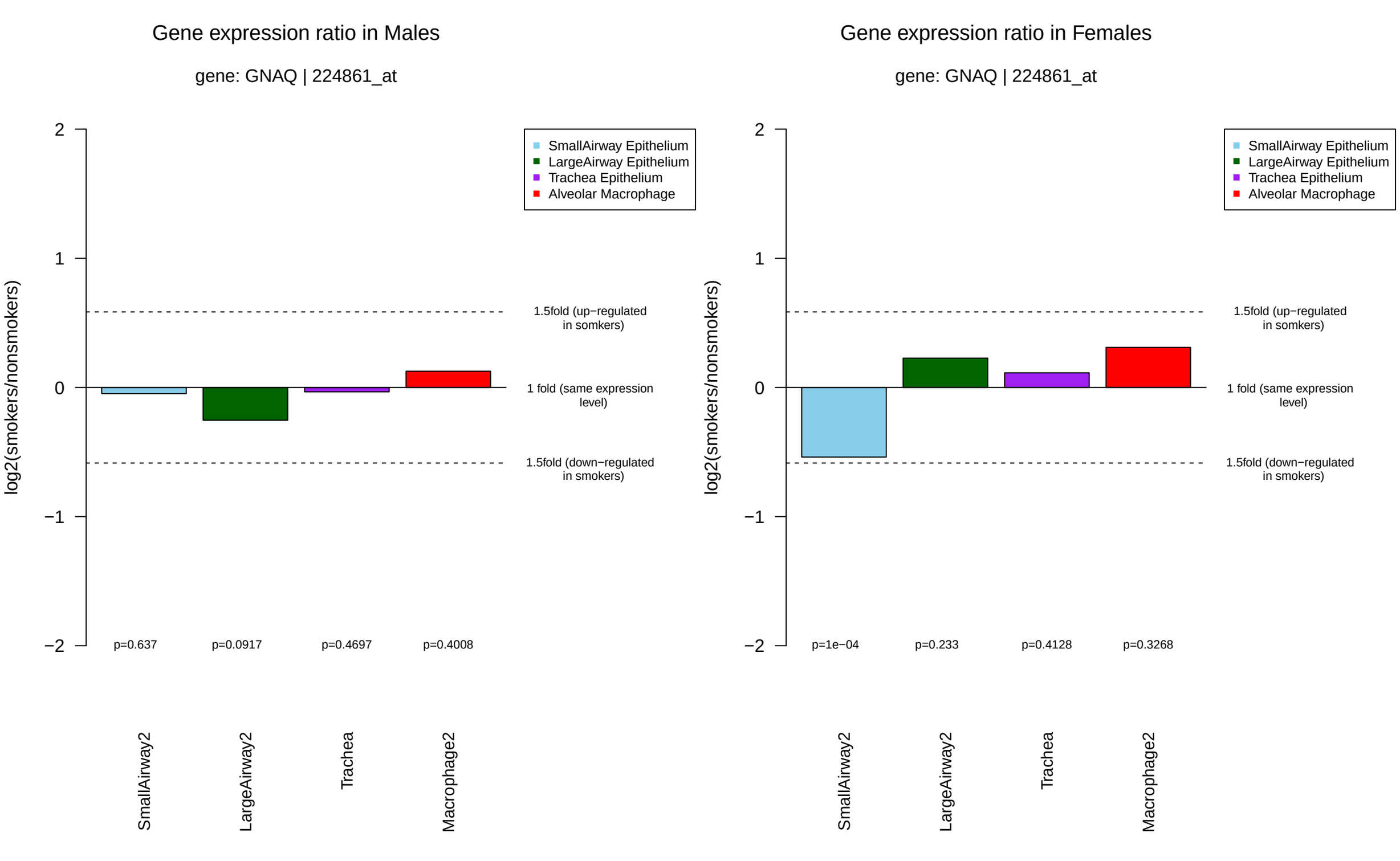
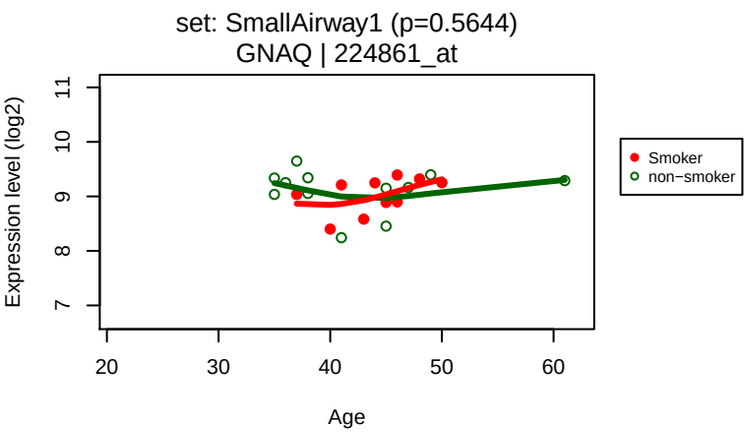
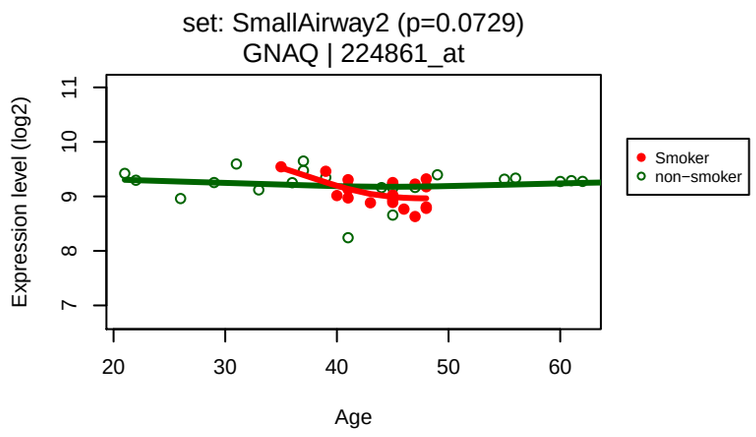
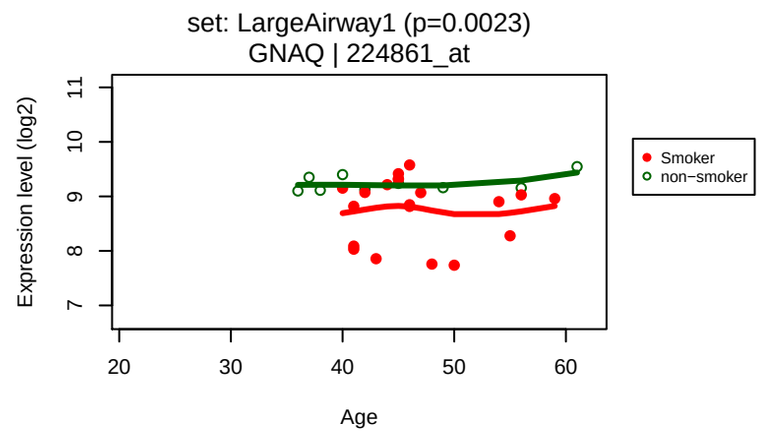

Gene expression of smokers and nonsmokers
gene: GNAQ | 224861_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5644
p=0.0729
p=0.0152
p=0.0023
p=0.5078
p=0.8195
p=0.1568
p=0.1426
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

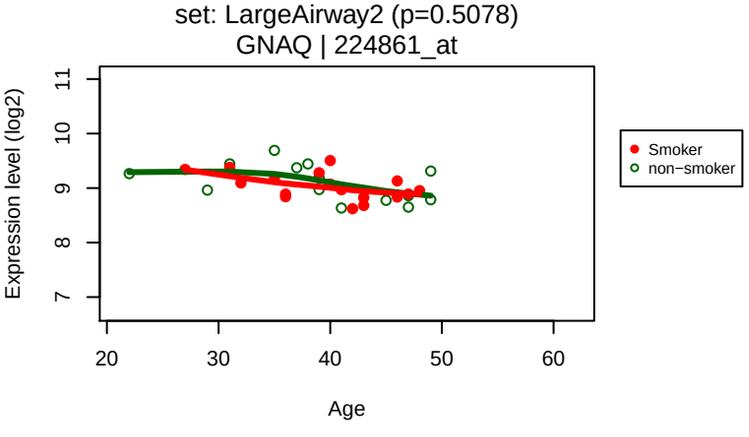

Gene expression ratio (smokers/nonsmokers)
gene: GNAQ | 224861_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5644
p=0.0729
p=0.0152
p=0.0023
p=0.5078
p=0.8195
p=0.1568
p=0.1426
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

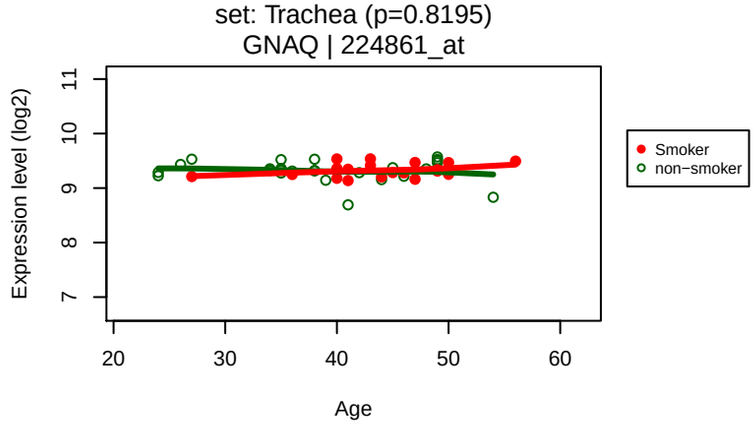

Gene expression ratio in Males
gene: GNAQ | 224861_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.637
p=0.0917
p=0.4697
p=0.4008
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAQ | 224861_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.233
p=0.4128
p=0.3268
SmallAirway2
LargeAirway2
Trachea
Macrophage2

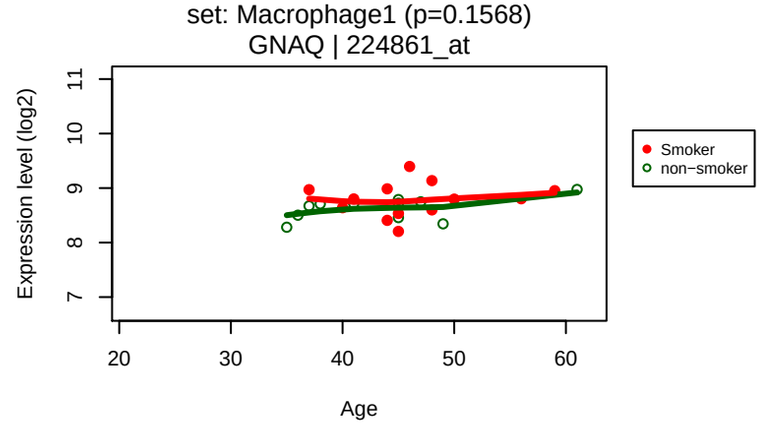

set: SmallAirway1 (p=0.5644)
GNAQ | 224861_at
Expression level (log2)
Age
Smoker
non−smoker

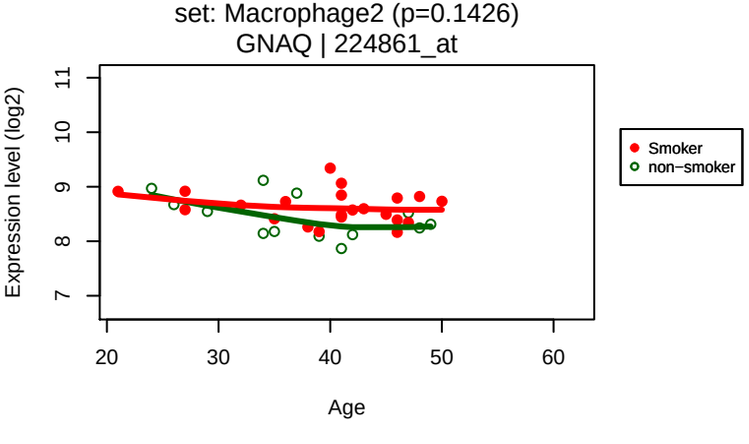

set: SmallAirway2 (p=0.0729)
GNAQ | 224861_at
Expression level (log2)
Age
Smoker
non−smoker

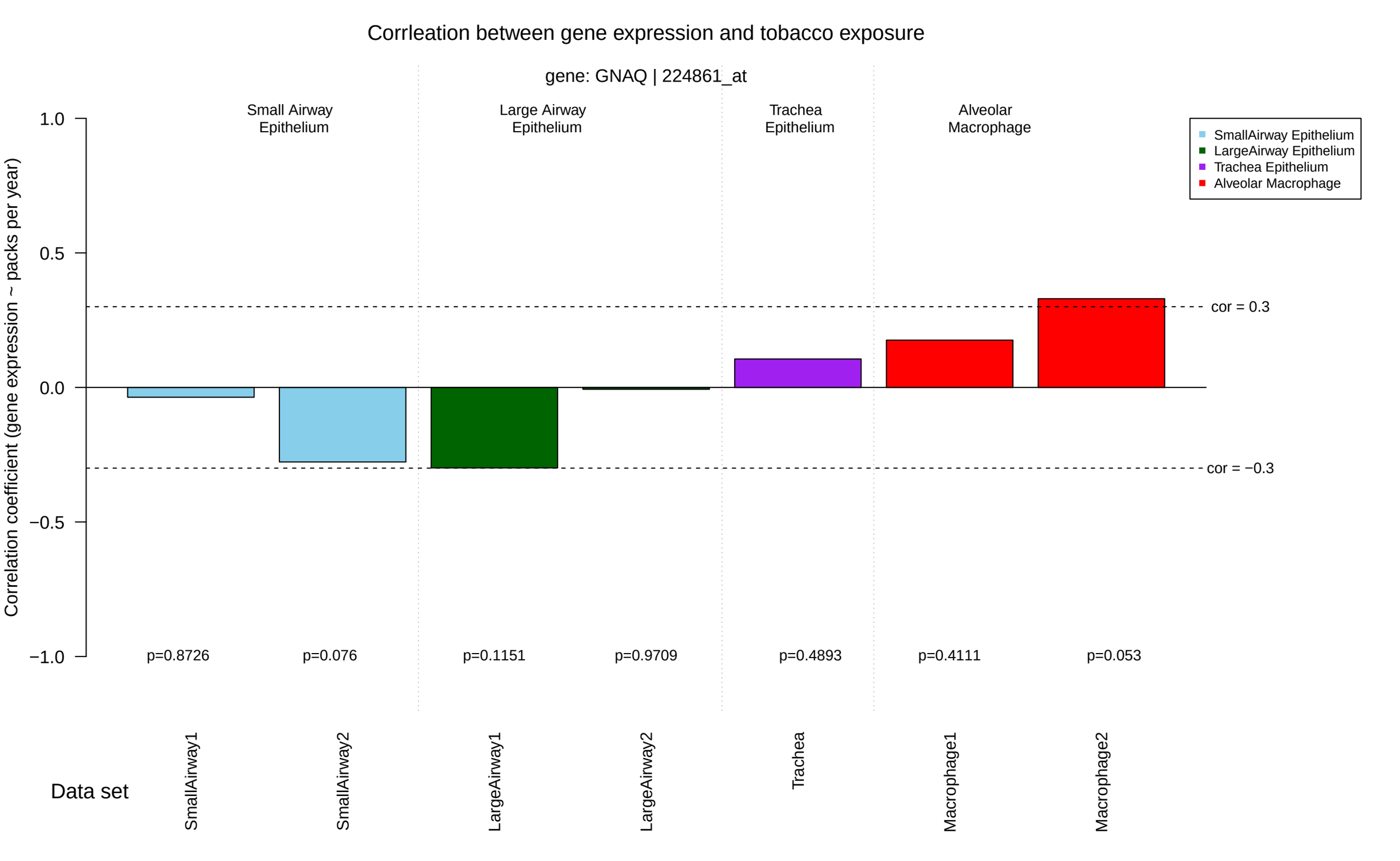

set: LargeAirway1 (p=0.0023)
GNAQ | 224861_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5078)
GNAQ | 224861_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8195)
GNAQ | 224861_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1568)
GNAQ | 224861_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1426)
GNAQ | 224861_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAQ | 224861_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8726
p=0.076
p=0.1151
p=0.9709
p=0.4893
p=0.4111
p=0.053
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2